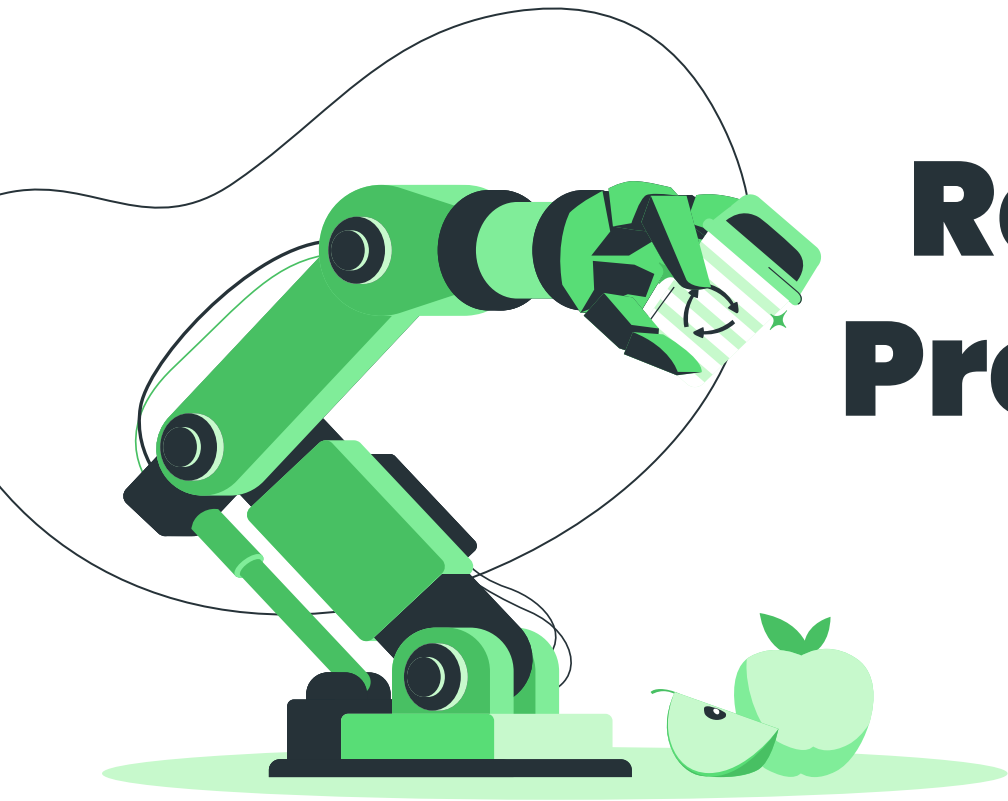


Litter Recognition Presentation

By Finn, Luke, Dev, and Ben



Our Client & Vision

- Moss & Gund Ltd. – actually just an individual.
- Scope Refinement.
- Our Vision:
 - Litter Detection
 - Litter Classification
 - Litter Collection

MVP



Robotics



35%

Manually controlled robot arm to collect litter.

Litter Detection



50%

Deep learning model to detect litter.

Controller



50%

Code to connect everything together.

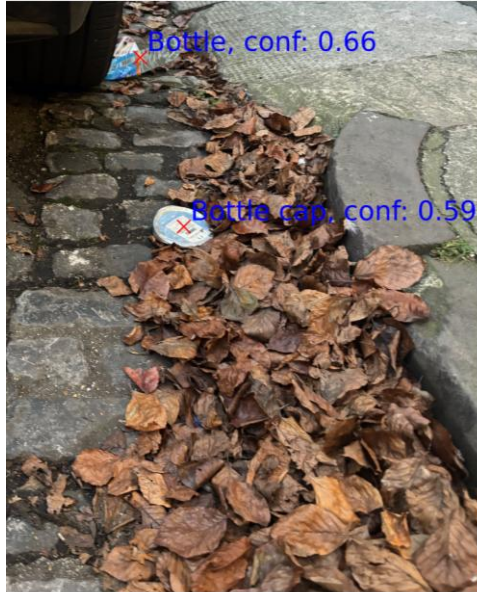
Stereo Vision



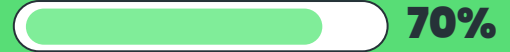
70%

Code to extract depth from dual-cameras.

Beta

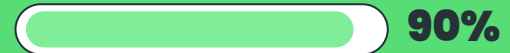


Robotics



Smoother movement + full range of motion.

Classification



Classification model to classify litter.

Controller



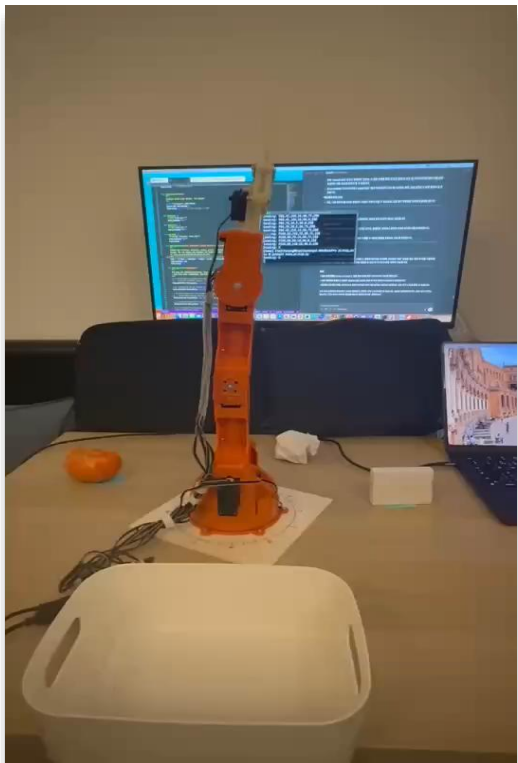
Asynchronous; the Raspberry Pi is very slow.

User Interface



Wireframes created and basic interface integrated with new logging system.

Final Release



Robotics

100%

New cameras, new mount.

Optimisations

100%

Model speeds, camera calibration, movement.

Controller

100%

Everything successfully connected together!

User Interface

100%

UI completed and fully integrated with the system.

Tech Stack

Main Components

Deep Learning, Stereo Vision, Robotics, Controller, User Interface

Languages

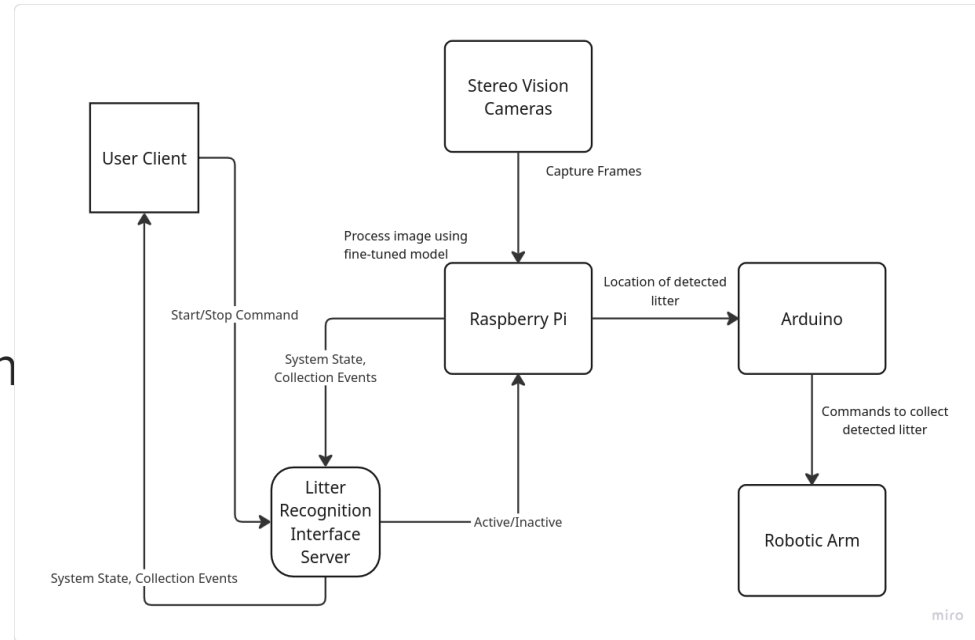
Python and C++ (Arduino)

Software PyTorch

PyTorch, YOLO, OpenCV, AWS

Hardware

Raspberry Pi, Arduino, Robot Arm, Dual Cameras



A Few User Stories

1

As an operator...

I want the system to detect and collect litter with 85%+ accuracy to reduce manual intervention.

2

As an operator...

I want to be able to easily set up and configure a new robot through a user interface so that new robots can be added to clean larger areas.

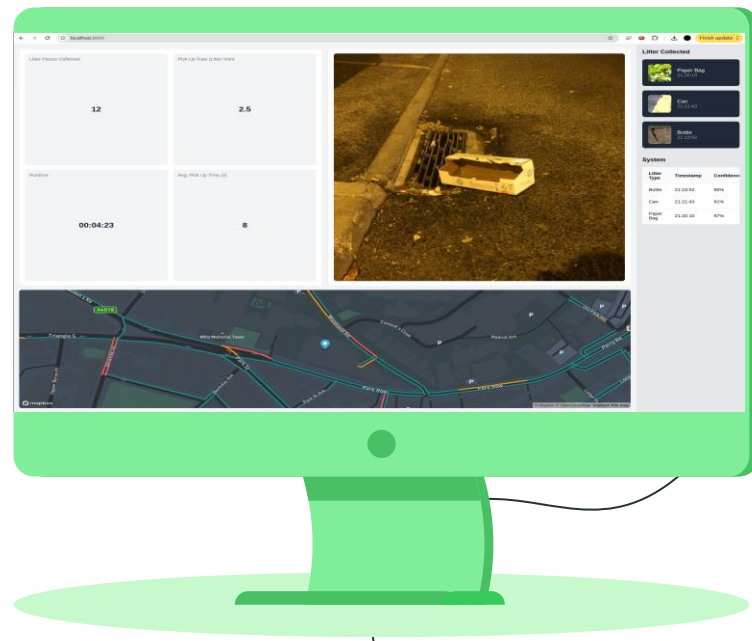
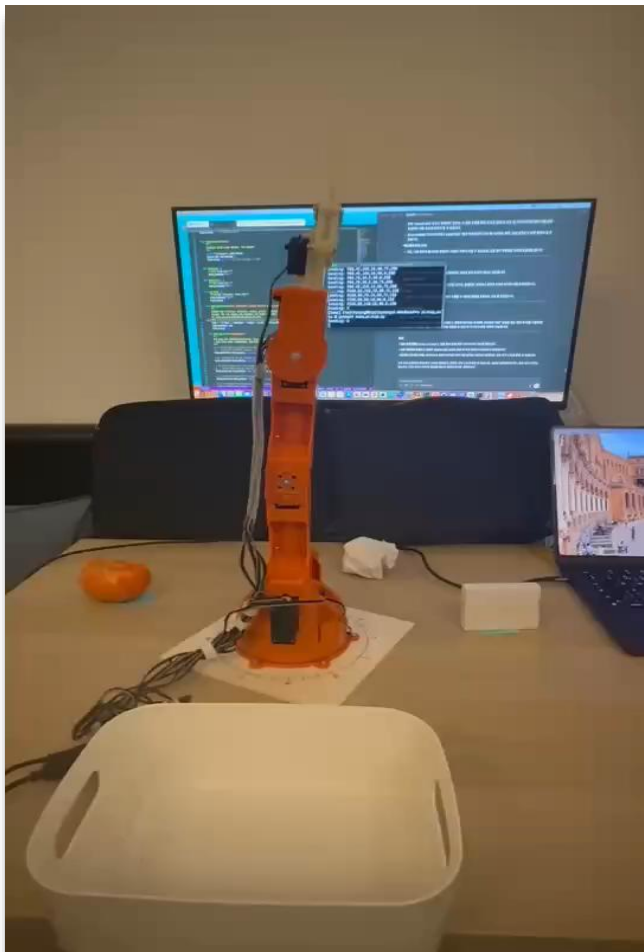
3

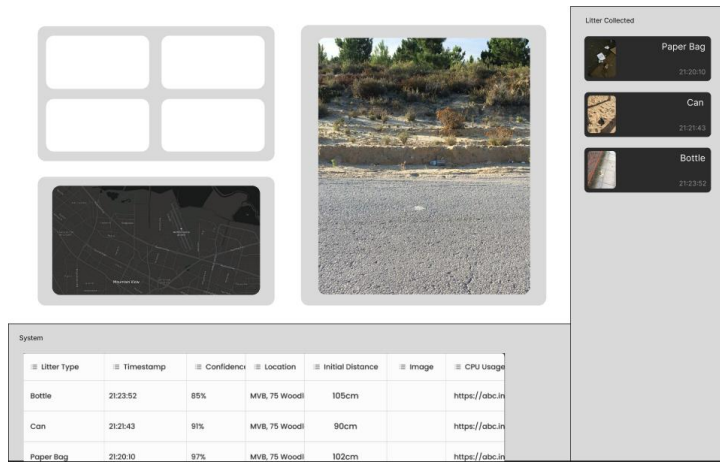
As a developer...

I want the User Interface to display various data and images live so I can analyse its performance.

I want to be able to annotate new litter so that I can improve detection accuracy of litter in new environments.

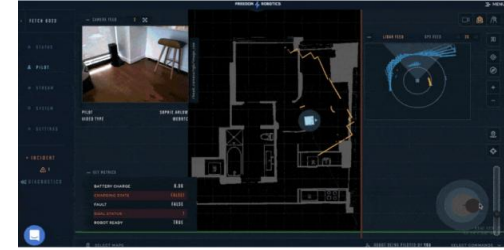
It's time for a Demo!





Initial Wireframe

User Interface



Robotics Interfaces for Inspiration

Initial User Story

I want the User Interface to display various data and images live so I can analyse its performance.

Additional Requirements

I want the User Interface to facilitate setup and control of robots.

Key Features

- Visibility of system state (live feed, location, and collections)
- Control of active/inactive
- New robot setup and activation
- Ability to monitor fleet of robots

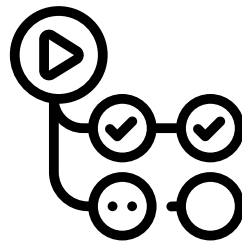
Development & Project Management

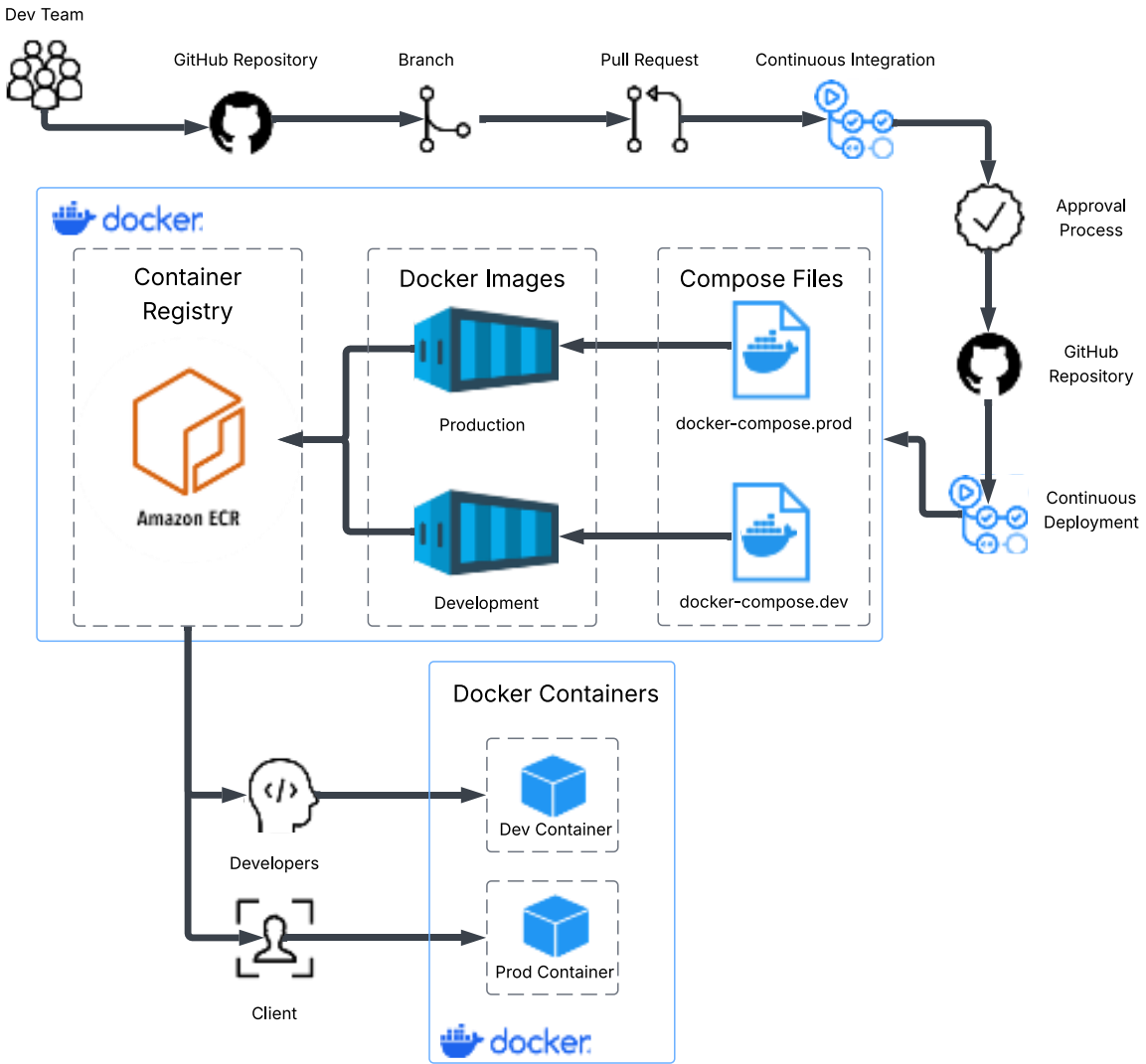
Tools Used:

- Issue and PR Templates.
- GitHub Actions for CI:
 - Unit, Partial, and Full Integration Tests.
 - Linting (Flake8).
 - Upgraded to report on code coverage: 66%.
- Dependabot.
- Docker + AWS ECR for CD.



unittest





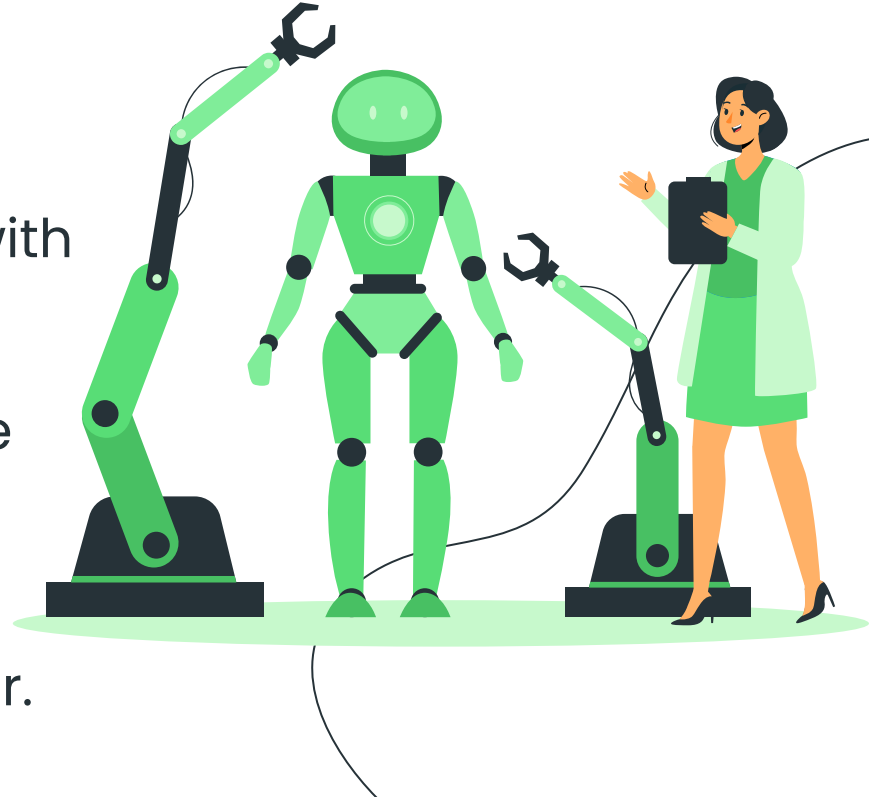
CI/CD

- Docker.
- Development and Production.
- Production requires hardware.
- Containers especially useful for this project.

Development & Project Management

Workflow:

- 2 Week Sprints + Sprint Reviews with the Client.
- Weekly meetings + stand ups in person (pair programming where possible).
- Agile Ceremonies.
- Frequent online meetings.
- Teams and Discord used together.



Key Challenges & Problems

Communication

- Discord wasn't effective.
- Late to meetings.
- Slow to review PRs – often waved through.
- Sections of the project were too isolated.

Hardware!

Pi cameras are a massive pain



Unrealistic Requirements

Dual robot arms!

Getting Started

Distributing workload for very unfamiliar tasks

Testing Day

4. The following two questions are about the User Interface and control of the robot. Please indicate how much you agree or disagree with the following statements

● Strongly disagree ● Disagree ● Neither agree nor disagree ● Agree ● Strongly agree

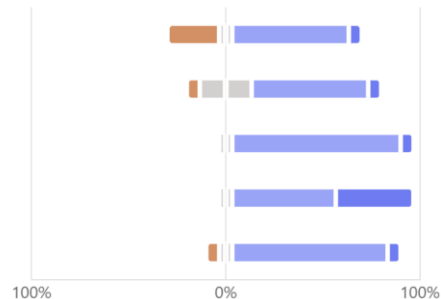
I find the robot's user interface intuitive and easy to use.

The information currently displayed on the UI is clear and useful.

The UI should provide real-time feedback on the robot's actions and decisions.

I would like the ability to manually override the robot's actions when necessary.

I believe that a graphical representation of the statistics is important.



3. Please indicate how much you agree or disagree with the following statements (NARS Inspired)

● Strongly disagree ● Disagree ● Neither agree nor disagree ● Agree ● Strongly agree

I feel uneasy knowing that an autonomous robot is operating in my surroundings.

I worry that the litter collection robot may malfunction and cause disruptions.

I feel uncomfortable with robots making independent decisions.

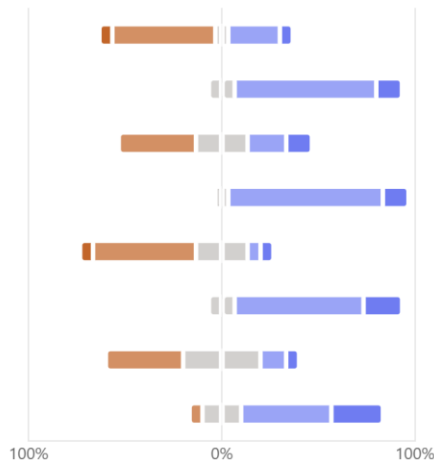
I am concerned that a litter collection robot may not recognize dangerous situations.

I believe that robots should not be used in tasks that involve interacting with public spaces.

I worry that the robot may not be able to differentiate between litter and important objects (e.g., personal belongings).

I think that society is becoming too reliant on robots and automation.

I am concerned about privacy issues related to robots collecting data while performing their tasks.



- Positive feedback!
- Safety and precision.
- Updated UI based on feedback.

Client Feedback

1

MVP

- Detection accuracy is good but could be improved.
- UI should display additional data.
- Needs a logging system.

2

Beta

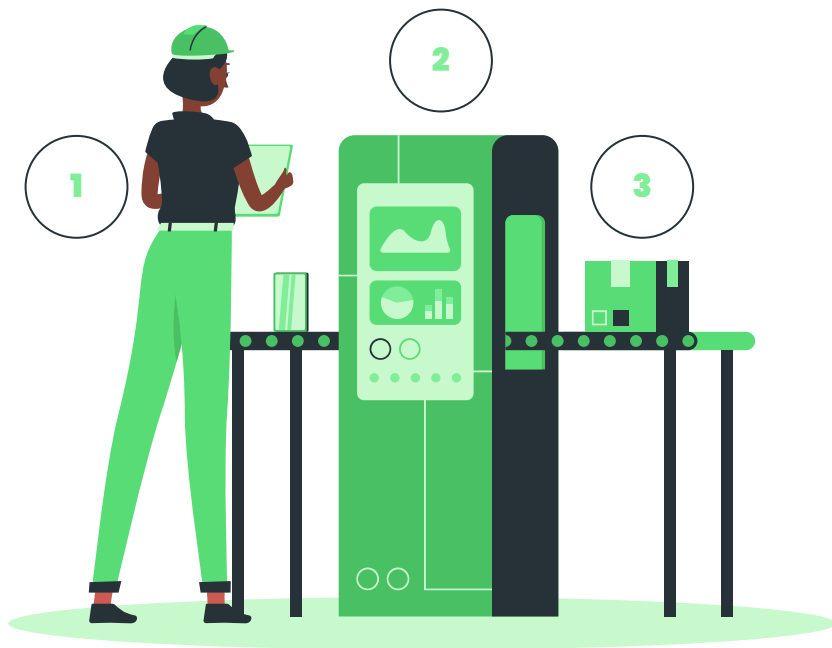
- Pleased with the progress.
- Displeased with biweekly reviews.
- Displeased about disconnected system.

3

Final

- Very pleased with final product.
- Very pleased with turn around of biweekly reviews.
- No changes needed but could use OKR's in the future.

Final Evaluation



1

The project was a success!

Despite numerous challenges, we created a working proof of concept.

2

What we've learnt

- Robotics is hard.
- Effective team collaboration.
- Wide variety of techniques.

3

What we would do differently

- Work more closely together; each section was isolated.
- Meet up in person more (running code was often impossible).
- Use a better robot arm.
- Never use Pi cameras again.



Thank you!

Any questions?